

31. The support of Fuzzy Set A is the set of all points  $x$  in  $X$  (is the universe of discourse) such that

Select one.

- ☒ a. Value of membership function of  $x > 0$
- ☐ b. Value of membership function of  $x = 0$
- ☐ c. Value of membership function of  $x = 0.5$
- ☐ d. Value of membership function of  $x = 1$



14 ID3 (one of the decision tree algorithms) makes use of information gain to arrange the attributes in a tree

Select one:

- ☒ a. True
- ☐ b. False



ID3 (one of the decision tree .14 algorithms) makes use of information gain to arrange the attributes in a tree

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a. False ☐

b. True ☐

Suppose a fuzzy set Young is defined as follows:

$$\text{Young} = \{(10, 0.5), (20, 0.8), (30, 0.8), (40, 0.5), (50, 0.3)\}$$

Then the crisp value of Young using Mean of maxima (MOM) defuzzification method is

Select one:

- ☐ a. 50
- ☐ b. 35
- ☐ c. 20
- ☒ d. 25



Three companies A, B and C supply 25%, 35% and 40% of the notebooks to a school. Past 2% of the notebooks produced by these companies are defective. If a notebook was found defective, the probability that the notebook was supplied by A?

Hint: Let D be the event that notebooks are defective then  $P(A|D)$ ?

Select one:

☐ a.  $\frac{13}{24}$

☐ b.  $\frac{44}{69}$

☐ c.  $\frac{11}{24}$

☒ d.  $\frac{25}{69}$



Transfer learning is:

Select one:

- ☐ a. an area of machine learning concerned with how software agents ought to take actions in an environment in order to maximize the notion of cumulative reward.
- ☒ b. a machine learning method where a model developed for a task is reused as the starting point for a model on a second task.
- ☐ c. a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labeled responses.
- ☐ d. the machine learning task of learning a function that maps an input to an output based on example input-output pairs.

Defuzzification is done to obtain .37

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a. Crisp output ☒

b. The best rule to follow ☐

c. Precise fuzzy value ☐

d. None of the above ☐

Suppose that Ali can decide to go to work by one of three modes of transportation, car, bus, or train. Because of high traffic, if he decides to go by car, there is a 40% chance he will be late. If he goes by bus, the probability of being late is 40%. The train is late with a probability of only 10 %.

One day Ali is late. Answer the following questions given that  $p(\text{car}) = 1/12$ ,  $p(\text{bus}) = 6/12$ ,  $p(\text{train}) = 5/12$ .

Hint: use  $h_{\text{MAP}} = \arg \max P(h|D) = \arg \max P(D|h) P(h)$

A. Compute  $p(\text{late}/\text{car}) \cdot p(\text{car})$ ? 0.033

B. Compute  $p(\text{late}/\text{train}) \cdot p(\text{train})$ ? 0.041

C. Compute  $p(\text{late}/\text{bus}) \cdot p(\text{bus})$ ? 0.2

D. What mode of transportation did he use? Bus

37. Defuzzification is done to obtain

Select one

- ☐ a. Crisp output
- ☐ b. None of the above
- ☒ c. Precise fuzzy value
- ☐ d. The best rule to follow

Consider the training examples shown in the Table below for a binary classification problem.

Where  $a_1$ ,  $a_2$  and  $a_3$  are the attributes

| Instance | $a_1$ | $a_2$ | $a_3$ | Target Class |
|----------|-------|-------|-------|--------------|
| 1        | T     | T     | 1.0   | +            |
| 2        | T     | T     | 6.0   | +            |
| 3        | T     | F     | 5.0   | -            |
| 4        | F     | F     | 4.0   | +            |
| 5        | F     | T     | 7.0   | -            |
| 6        | F     | T     | 3.0   | -            |
| 7        | F     | F     | 8.0   | -            |
| 8        | T     | F     | 7.0   | +            |
| 9        | F     | T     | 5.0   | -            |

A. What is the entropy of this collection of training examples? 0.9911

B. What is the information gains of  $a_2$  relative to these training examples? 0.1751

What is the name of this phenomenon that sometimes occurs in machine learning model. "occurs when the training error of the learnt model is small but the test error is large"?

Select one:

- ☐ a. Classification
- ☒ b. Overfitting
- ☐ c. Generalization
- ☐ d. Underfitting



44. What Is The Purpose of Aggregation (in Fuzzy Inference System)?

Select one:

- ☐ a. To gather all the different fuzzy set outputs and average them out to get a single value
- ☐ b. To gather all the possible inputs and use the average to gain an output
- ☒ c. To gather all the different fuzzy set outputs and combine them into a single fuzzy set outputs
- ☐ d. To subtract all the output fuzzy set values from the input values

Which of the following cannot be stated  
?using fuzzy logic

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a. Height of a person ☐

b. Speed of a car ☐

c. Color of an apple ☐

d. Date of birth of a student ☒

30. A fuzzy set A is closed if:

Select one:

- ☐ a. Limit of both sides of membership function are 1
- ☒ b. Limit of both sides of membership function are 0
- ☐ c. Limit of infinity side of membership function is 1, the other limit is 0.
- ☐ d. Limit of infinity side of membership function is 0, the other limit is 1.

Which statement is wrong related to a  
SparX? decision tree



We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats.

The identification result of the model is that 140 images contain cats and 60 do not contain cats.

Given that TP (true positive) of the model = 135.

What is the Accuracy of the model?

Select one:

- ☐ a. 75 %
- ☐ b. 82.5 %
- ☒ c. 96.4 %
- ☐ d. 75.7 %



Previous probabilities in Bayes Theorem that are changed with help of new available information are classified as:

Select one:

- ☐ a. dependent probabilities
- ☒ b. posterior probabilities
- ☐ c. independent probabilities
- ☐ d. interior probabilities



What is the subject of artificial intelligence?

Select one:

- ☐ a. Linguistics
- ☒ b. Comprehensive interdisciplinary and marginal disciplines
- ☐ c. Mathematics and Physiology
- ☐ d. Psychology and Physiology

In supervised learning, training set of data includes

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a. Output ☐

b. None ☐

c. Input ☐

d. Both input and output ☒

What are the applications of Fuzzy Inference Systems?

Select one:

- ☐ a. Simulink, boiler and CD recording
- ☐ b. Restrict power usage, telephone lines and sort data
- ☐ c. Automatic control, decision analysis and data classification
- ☐ d. Wireless services, heat control and printers

# What are the applications of Fuzzy Inference Systems?

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In machine learning models, the data :validation set is used to tune

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a. overfitting ☐

b. underfitting ☐

c. The model internal parameters ☐

d. The model hyper parameters ☐

What is the name of this principle in machine learning: "The goal of machine learning is that the learnt model should perform well on new samples, not just those on which the model has been trained"

Select one:

- ☐ a. Underfitting
- ☐ b. Classification
- ☐ c. Overfitting
- ☒ d. Generalization



:In Naive Bayes classification

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- a. Assumption that all input attributes are conditionally dependent ☐
- b. Assumption that all input attributes are related to each other ☐
- c. Assumption that all input attributes are mutually dependent ☐
- d. Assumption that all input attributes are conditionally independent ☒

Which statement is wrong related to a  
?decision tree

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- a. A decision tree is a tree-like ☐ structure (a binary tree or a non-binary tree).
- b. Each branch stores the entropy of ☐ attribute.
- c. Each non-leaf node represents a test ☐ on a feature attribute.
- d. Each leaf node stores a class ☐.

What is a membership function of a fuzzy  
?set

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- a. none ☐
- b. It is a set of elements that have some common property (such as 'Hot') ☐
- c. A collection of fuzzy sets ☐
- d. It describes the degree of membership that an object belongs to the set ☐

Two boxes containing candies are placed on a table. The boxes are labelled B1 and B2. Box B1 contains 6 cinnamon candies and 5 ginger candies. Box B2 contains 4 cinnamon candies and 7 pepper candies.

The boxes are arranged so that the probability of selecting box B1 is  $\frac{1}{4}$  and the probability of selecting box B2 is  $\frac{3}{4}$ . Ahmad is blindfolded and asked to select a candy. He will win a mobile phone if he selects a cinnamon candy.

If he wins a mobile phone, what is the probability that the candy was from the first box?

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a.  $\frac{6}{33}$  ☐

b.  $\frac{7}{13}$  ☐

c.  $\frac{6}{18}$  ☐

d.  $\frac{6}{44}$  ☐

Consider two fuzzy subsets of the set .1

.X,  $X = \{a, b, c, d, e\}$  referred to as A and B

$$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$$

$$\text{and } B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$$

? = Find  $A \cup B$

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a.  $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$  ☐

b.  $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$  ☐

c.  $\{1/a, 0.9/b, 0.2/c, 0.8/d, 0.2/e\}$  ☐

d.  $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$  ☐

|    |     |     |     |     |
|----|-----|-----|-----|-----|
| 3  | YES | YES | NO  | NO  |
| 4  | YES | NO  | YES | YES |
| 5  | YES | YES | YES | YES |
| 6  | NO  | YES | NO  | NO  |
| 7  | YES | NO  | YES | YES |
| 8  | YES | NO  | YES | YES |
| 9  | NO  | YES | YES | YES |
| 10 | YES | YES | NO  | YES |
| 11 | NO  | YES | NO  | NO  |
| 12 | NO  | YES | YES | NO  |
| 13 | NO  | YES | YES | NO  |
| 14 | YES | YES | NO  | NO  |

Using ID3 algorithm compute  $IG(S, \text{Fever})$   
 ?(Information gain of fever)

Hint: Final answer should contain exactly 2  
 خانتيں عشریتين).digits after decimal point  
 فقط)

الجواب:

## Question

Fuzzification is done to obtain  
Select one:

- a. Crisp output
- b. None of the above
- c. The best rule to follow
- d. Precise fuzzy value

:In Naive Bayes classification

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- a. Assumption that all input attributes ☐ are related to each other
- b. Assumption that all input attributes ☐ are conditionally dependent
- c. Assumption that all input attributes ☐ are conditionally independent
- d. Assumption that all input attributes ☐ are mutually dependent

Two fuzzy sets A and B with membership functions  $G(x)$  and  $H(x)$ , respectively defined as below

.A = Hot Climate with  $G(x)$  as the MF

.B = Cold Climate with  $H(x)$  as the M.F

:Pleasant (best ) climate is given by

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Spa  
POLYTECHNIC

a.  $1 - G(x)$  ☐

b.  $1 - H(X)$  ☐

c.  $\min (G(x), H(x))$  ☐

d.  $\max (G(x), H(x))$  ☐

What is the name of this phenomenon that sometimes occurs in machine learning model: "occurs when the training error of the learnt model is small but the test error?" is large

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- a. Overfitting ☐
- b. Underfitting ☐
- c. Generalization ☐
- d. Classification ☐

## سؤال 16

غير مجاب عليه بعد

الدرجة من 1.00

علم هذا السؤال 

What is the Bandwidth of fuzzy set A ? The  
:set is given as follow

$A = (10, 0.1), (15, 0.2), (20, 0.5), (25, 0.4),$   
 $(30, 0.4), (35, 0.5), (40, 0.2), (45, 0.1)$

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a. 35 ☐

b. 20 ☐

c. -15 ☐

d. 15 ☐

Consider two fuzzy subsets of the set  $X$ ,  $X = \{a, b, c, d, e\}$  referred to as  $A$  and  $B$

$$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$$

$$\text{and } B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$$

SPAR POLYTECHNIC ? = Find A  
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- a.  $\{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$  ☐
- b.  $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$  ☐
- c.  $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$  ☐
- d.  $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$  ☐

سؤال 14

غير مجاب عليه بعد

الدرجة من 1.00

علم هذا السؤال

An example of a hyperparameter of a  
neural network

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- a. Weights ☐
- b. Number of inputs ☐
- c. Learning rate ☐
- d. Bias ☐

If  $x$  is  $A$  then  $y$  is  $B$  else  $y$  is  $C$ . The output of the given fuzzy rule is

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a. a crisp set ☐

b. a fuzzy relation ☐

c. a membership function ☐

d. a fuzzy set ☐

Two boxes containing candies are placed on a table. The boxes are labelled B1 and B2. Box B1 contains 6 cinnamon candies and 5 ginger candies. Box B2 contains 4 cinnamon candies and 7 pepper candies.

The boxes are arranged so that the probability of selecting box B1 is  $\frac{1}{4}$  and the probability of selecting box B2 is  $\frac{3}{4}$ . Ahmad is blindfolded and asked to select a candy. He will win a mobile phone if he selects a cinnamon candy.

If he wins a mobile phone, what is the probability that the candy was from the first box?

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a.  $\frac{6}{44}$  ☐

b.  $\frac{6}{33}$  ☐

c.  $\frac{6}{18}$  ☐

d.  $\frac{7}{13}$  ☐

1. Consider two fuzzy subsets of the set  $X$ ,  $X = \{a, b, c, d, e\}$  referred to as A and B

$$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$$

$$\text{and } B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$$

SPARK  
POLYTECHNIC  
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? = Find  $A \cup B$

a.  $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$  ☐

b.  $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$  ☐

c.  $\{1/a, 0.9/b, 0.2/c, 0.8/d, 0.2/e\}$  ☐

d.  $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$  ☐

Previous probabilities in Bayes Theorem that are changed with help of new available information are classified as

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a. posterior probabilities ☐

b. dependent probabilities ☐

c. independent probabilities ☐

d. interior probabilities ☐

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الدرجة من 1.00

علم هذا السؤال

Which of the following cannot be stated  
?using fuzzy logic

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a. Speed of a car ☐

b. Height of a person ☐

c. Color of an apple ☐

d. Date of birth of a student ☐

## سؤال 8

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الدرجة من 1.00

علم هذا السؤال 

Decision tree is an algorithm for which  
?machine learning task

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a. regression ☐

b. dimensionality reduction ☐

c. clustering ☐

d. classification ☐

## سؤال 7

غير مجاب عليه بعد

الدرجة من 1.00

علم هذا السؤال

ID3 (one of the decision tree algorithms) makes use of information gain to arrange the attributes in a tree

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a. True ☐

b. False ☐



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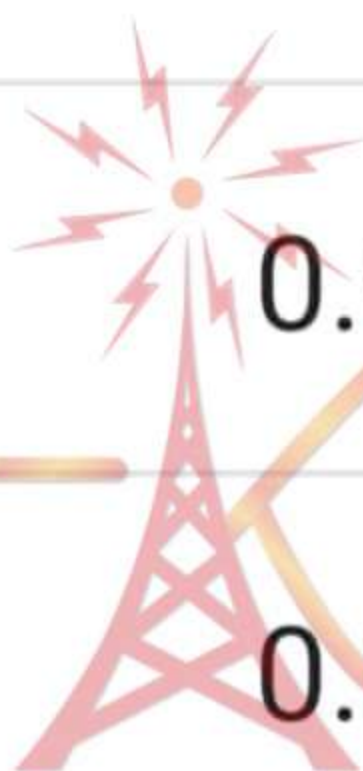
0.1751



0.9839



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0.0771



.0.0072



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0.9711



0.9811

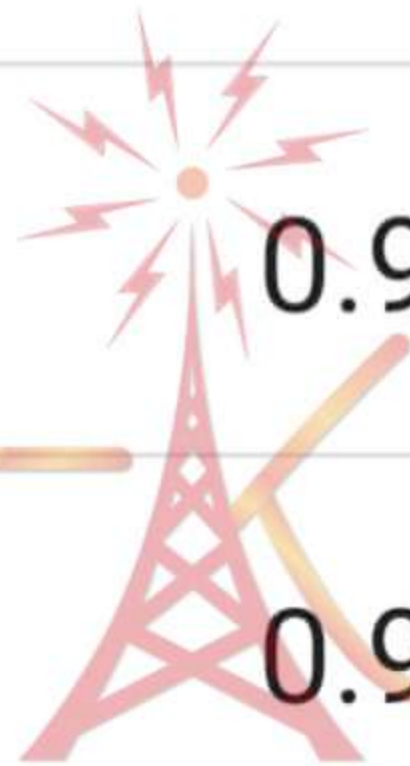


0.9111



0.9911

Spar  
POLYTECHNIC



Consider the training examples shown in the Table below for a binary classification problem.

Where  $a_1$ ,  $a_2$  and  $a_3$  are the attributes

| Instance | $a_1$ | $a_2$ | $a_3$ | Target Class |
|----------|-------|-------|-------|--------------|
| 1        | T     | T     | 1.0   | +            |
| 2        | T     | T     | 6.0   | +            |
| 3        | T     | F     | 5.0   | -            |
| 4        | F     | F     | 4.0   | +            |
| 5        | F     | T     | 7.0   | -            |
| 6        | F     | T     | 3.0   | -            |
| 7        | F     | F     | 8.0   | -            |
| 8        | T     | F     | 7.0   | +            |
| 9        | F     | T     | 5.0   | -            |

A. What is the entropy of this collection of

✓ اختر... training examples

B. What is the information gains of  $a_2$  ?relative to these training examples

✓ اختر...

## سؤال 5

غير مجاب عليه بعد

الدرجة من 1.00

علم هذا السؤال 

Regarding entropy which statement is  
.False

Hint: S is a collection of positive and  
(negative examples

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- a. Entropy is 1 when S contains an ☐ equal number of positive and negative examples
- b. Entropy is 0 when S contains an ☐ equal number of positive and negative examples
- c. Entropy is measured in bits ☐
- d. Entropy measures the uncertainty ☐

We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats. The identification result of the model is that 140 images contain cats and 60 do not contain cats. Given that TP (true positive) of the model = 135

?What is the Accuracy of the model

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% a. 96.4 ☐

% b. 75.7 ☐

% c. 82.5 ☐

% d. 75 ☐

Suppose that Ali can decide to go to work by one of three modes of transportation, car, bus, or train. Because of high traffic, if



0.2

1/12

0.3

0.4



Suppose that Ali can decide to go to work by one of three modes of transportation, car, bus, or train. Because of high traffic, if



0.1

0.2

0.062

0.033



0.4



0.025



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0.1



0.041

Suppose that Ali can decide to go to work by one of three modes of transportation, car, bus, or train. Because of high traffic, if he decides to go by car, there is a 60 % chance he will be late. If he goes by bus, the probability of being late is 30%. The % train is late with a probability of only 15

One day Ali is late. Answer the following questions given that  $p(\text{car})=6/12$ ,  $p(\text{bus})=1/12$   $p(\text{train}) = 5/12$

Hint: use  $h_{\text{MAP}} = \arg \max P(h/D) = \arg \max P(D/h) P(h)$



?A. Compute  $p(\text{late}/\text{car}) * p(\text{car})$

?B. Compute  $p(\text{late}/\text{train}) * p(\text{train})$



?C. Compute  $p(\text{late}/\text{bus}) * p(\text{bus})$

D. What mode of transportation did he



?use اختر...

In unsupervised learning, training set of data includes

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a. Input ☐

b. Output ☐

c. Both input and output ☐

d. None ☐

30. A fuzzy set A is closed if

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- a. Limit of infinity side of membership function is 0, the other limit is 1 ☐
- b. Limit of infinity side of membership function is 1, the other limit is 0 ☐
- c. Limit of both sides of membership function are 1 ☐
- d. Limit of both sides of membership function are 0 ☐

:Transfer learning is

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- a. an area of machine learning ☐ concerned with how software agents ought to take actions in an environment in order to maximize the notion of cumulative reward
- b. a type of machine learning algorithm ☐ used to draw inferences from datasets consisting of input data without labeled responses
- c. a machine learning method where a ☐ model developed for a task is reused as the starting point for a model on a second task
- d. the machine learning task of ☐ learning a function that maps an input to an output based on example input-output pairs